Lokal tarmoq asoslari

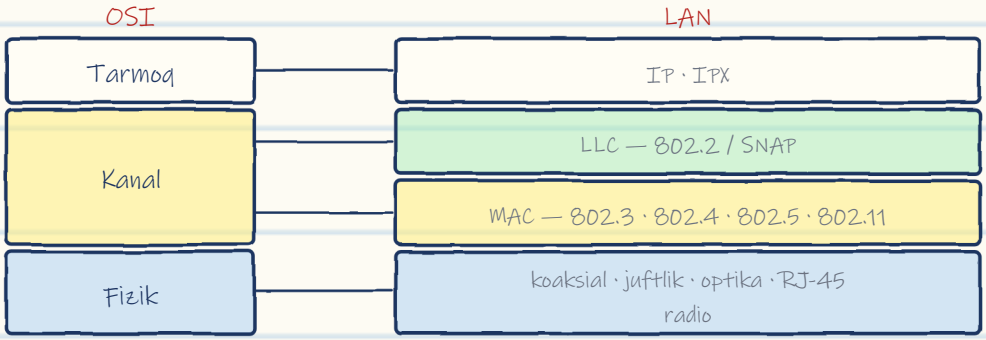
LAN Fundamentals

Asosiy atamalar

- LAN — bino yoki kampus ichidagi lokal tarmoq (1- va 2-qatlam)
- MAC adres — tarmoq kartasining 48 bitli apparat manzili
- Kadr (frame) — kanal qatlamining ma'lumot birligi
- CSMA/CD — tinglab uzatish, to'qnashuvni sezib qayta yuborish
- Kolliziya domeni — signallar to'qnashadigan umumiy hudud
- Hub · kommutator — signalni takrorlaydi · kadrni saqlab uzatadi

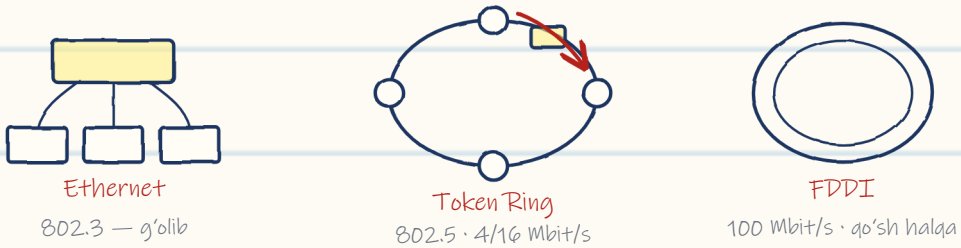
1. LAN va OSI modeli

LAN texnologiyalari faqat 1- va 2-qatlamga tegishli.



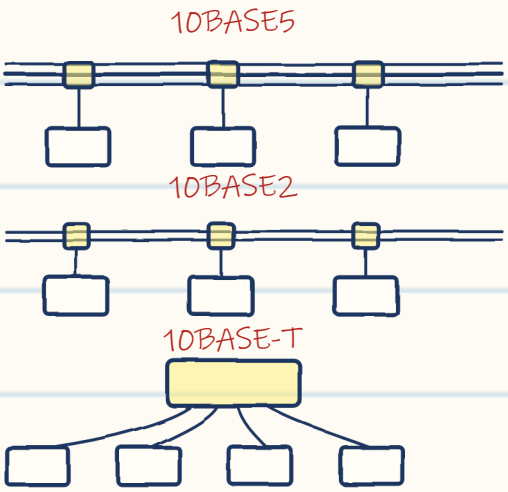
1-rasm. LLC hamma uchun bir xil, MAC — muhitga qarab

2. LAN texnologiyalari



2-rasm. Token — ruxsat belgisi; FDDI — optik qo'sh halqa

3. Erta Ethernet



yo'g'on koaksial · 500 m
transiver orqali ulanadi

ingichka koaksial · 200 m
BNC T-ulagich

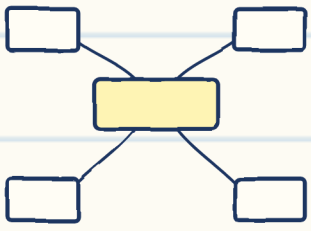
juftlik CAT-5 · RJ-45 · 100 m
yulduz · markazda hub
oson quriladi, nosozlik topiladi

3-rasm. Shina → yulduz: bitta uzilish endi hammani to'xtatmaydi

Nom	Tezlik	Muhit	Masofa
10BASE5	10 Mbit/s	yo'g'on koaksial	500 m
10BASE2	10 Mbit/s	ingichka koaksial	200 m
10BASE-T	10 Mbit/s	o'ralgan juftlik	100 m

4. Hub

hub — fizik qatlam

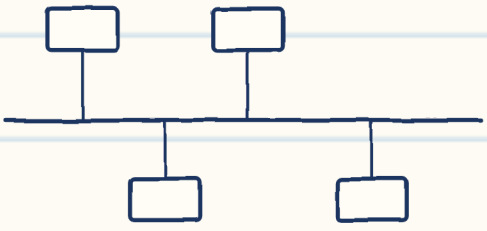


fizik: yulduz

teng

yarim dupleks

bir vaqtda faqat bittasi uzatadi



mantiqiy: shina

5. MDI/MDIX va kabel

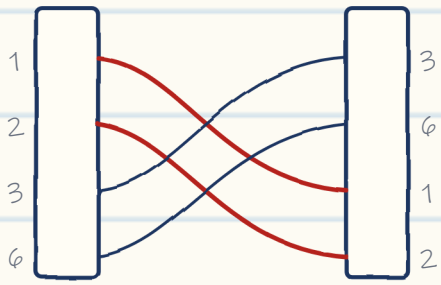
to'g'ri kabel



MDI — MDIX

PC — kommutator porti

kesishgan kabel



MDI — MDI

PC — PC · switch — switch

H3C kommutatorlarida auto MDI/MDIX

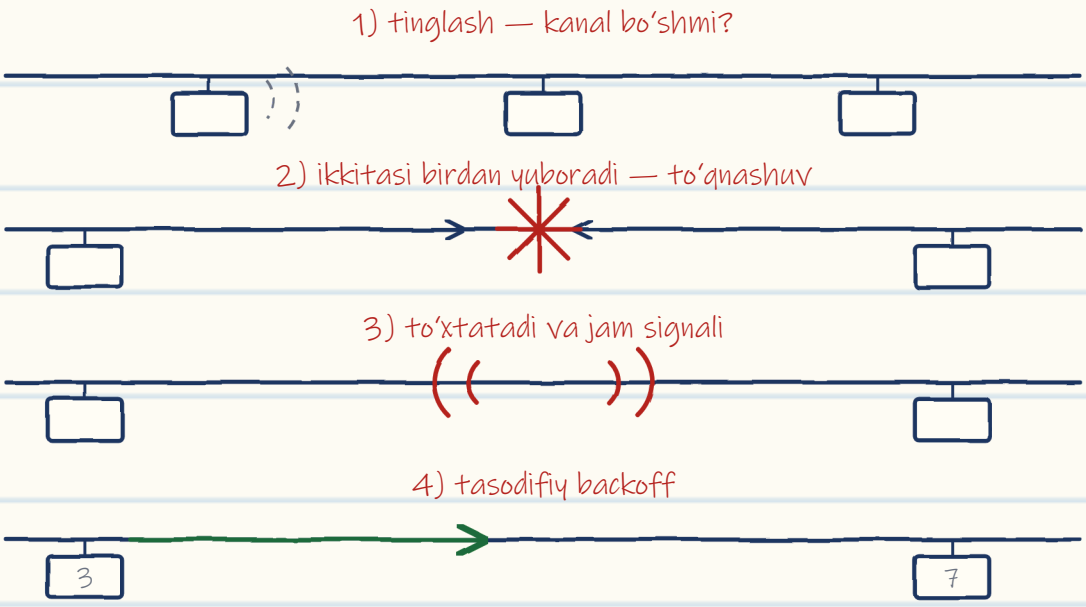
juftlik: 8 tomir = 4 juft, uchida RJ-45

4-rasm. Kesishganda 1–3 va 2–6 kontaktlar almashadi

to'g'ri	MDI — MDIX	PC—switch, router—switch
kesishgan	MDI — MDI	PC—PC, switch—switch
auto	port o'zi tanlaydi	kabel turi ahamiyatsiz

- UTP — arzon, o'rnatish oson, himoya kuchsiz
- STP — metall ekran, qimmat, EMI dan himoya
- kategoriya 3 · 4 · 5 · 5e · 6 — eng ko'p CAT-5

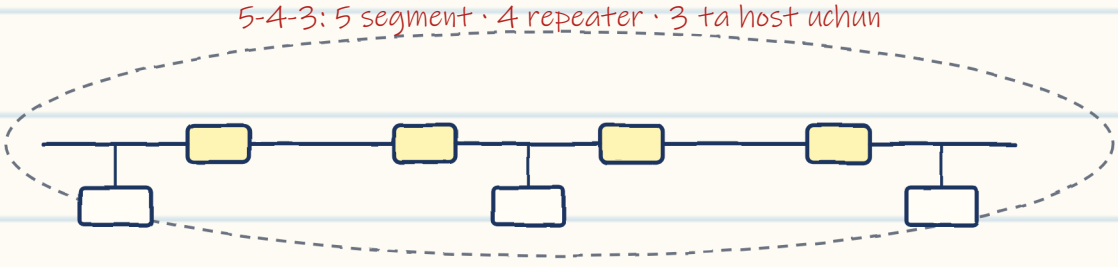
6. CSMA/CD



kichik backoff — birinchi yuboradi

5-rasm. Eshit → yubor → to'qnashuvni sez → kut → qayta yubor

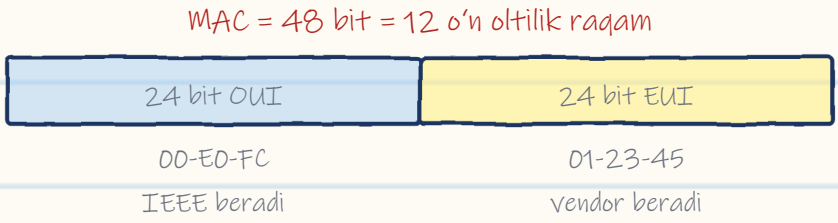
7. Kolliziya domeni

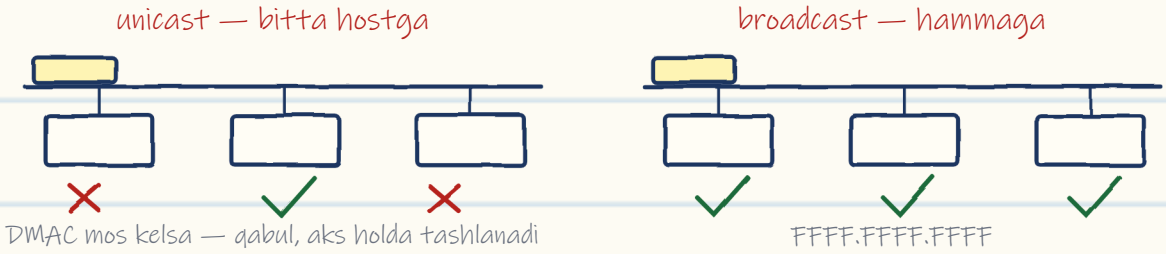


hammasi bitta kolliziya domeni

tavsiya: domenda 50 tagacha stansiya

8. MAC adres · unicast va broadcast





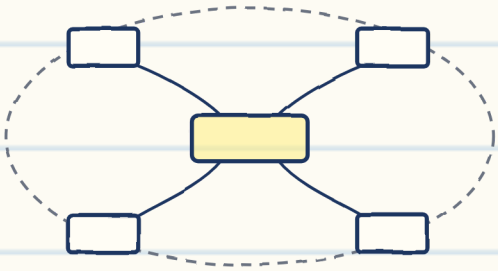
6-rasm. OUI — IEEE, EUI — ishlab chiqaruvchi

9. Tez Ethernet

Nom	Muhit	Masofa	Standart
100BASE-TX	2 juft CAT-5	100 m	802.3u
100BASE-FX	ko'p modli optika	2000 m	802.3u
1000BASE-SX	ko'p modli optika	275 / 550 m	802.3z
1000BASE-LX	bir modli optika	550 / 5000 m	802.3z
1000BASE-T	4 juft CAT-5	100 m	802.3ab

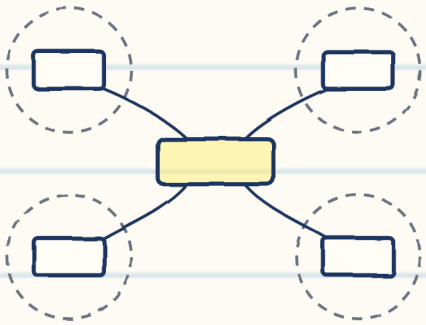
10. Hub va kommutator

hub — signalni takrorlaydi



1 kolliziya domeni

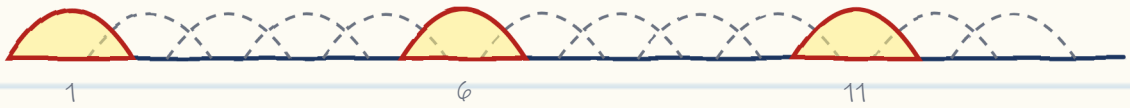
switch — kadrni saqlab uzatadi



har port — alohida domen

11. WLAN — 802.11

2,4 GHz · 13 kanal · har biri 22 MHz



kesishmaydigan kanallar: 1 · 6 · 11

CSMA/CA — to'qnashuvdan qochish

DIFS + tasodifiy backoff + MAC ACK

Xulosa

- LAN = OSI ning fizik va kanal qatlami; kanal = LLC + MAC
- Ethernet — 802.3, bugungi kunda yagona hukmron
- CSMA/CD: eshit · yubor · sez · kut · qayta yubor
- MAC — 48 bit; broadcast FFFF.FFFF.FFFF
- Switch har portni alohida kolliziya domeniga ajratadi

Tekshirish

1. 10BASE-T masofasi? — 100 m
2. MAC uzunligi va bo'linishi? — 48 bit = OUI 24 + EUI 24
3. WLAN da kesishmaydigan kanallar? — 1, 6, 11